Namma Metro Smart-Card Fare System

Github: https://github.com/fr-varun/DSA0103-c-.git

Assignment Report — Tasks 1–12

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TASK 1 – Object-Oriented Analysis (4 Marks)

1.1 Objects Identified

The fare problem, read through an OO lens, yields five clear objects. Each is justified below.

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1.1.1 SmartCard

Aspect	Detail
Why it is an object	It has persistent state (balance) that changes over time and rules that govern changes (minimum balance). The only way to keep that rule consistent is to bundle data and behaviour together.
Data	balance (long long, paise), cardId (string), ownerId (string)
Behaviour	recharge(amount), deduct(fare), canTravel(fare), getBalance()

The balance must *never* drop below Rs 50 (5000 paise) after a journey. Because this invariant belongs to the card, `deduct` enforces it internally — no external code can accidentally forget the check.

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1.1.2 Passenger

Aspect	Detail
Why it is an object	Different passengers get different concessions (student, senior citizen). Polymorphism lets each subclass override concessionRate() without changing the caller.
Data	name, category (enum: Regular, Student, Senior)

Behaviour	concessionRate() → double, getCategory()

Without a Passenger object, every fare-computation function would need an extra `int category` parameter and an `if-else` chain that must be duplicated everywhere.

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1.1.3 Journey

Aspect	Detail
Why it is an object	A journey captures a snapshot of one trip: its zone, hour, whether it was peak, and the fare paid. Storing journeys lets the system produce audit logs and monthly statements.
Data	zone (Zone enum), hour (int), isPeak (bool), farePaid (long long)
Behaviour	computeFare(), isValid()

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1.1.4 Zone

Aspect	Detail
Why it is an object / enum	A zone is a value type with a fixed, closed set of members. An enum class Zone makes it type-safe (you cannot pass an int where a Zone is expected) and self-documenting.
Data	Inner = 0, Middle = 1, Outer = 2
Behaviour	None — purely a discriminator; the fare table maps Zone → paise

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1.1.5 FareTable

Aspect	Detail
Why it is an object	All fare rules live in one place. When a fare changes, only

	FareTable is edited. Every other object queries it.
Data	BASE_PAISE[3], PEAK_SURCHARGE_PERCENT, MIN_BALANCE_PAISE
Behaviour	computeFare(zone, isPeak, passenger) → long long

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1.2 Procedural vs Object-Oriented Style

Procedural approach:

Global variables hold the balance and passenger category. Free functions like `computeFare(int zone, bool peak, int category)` and `deductBalance(long long& balance, long long fare)` implement the logic. Any function in the program can read or write the balance directly.

Risks of the procedural approach:

- The minimum-balance rule must be checked in *every* place that calls `deductBalance`. If one call site forgets, a negative balance can appear (exactly the bug mentioned in the problem statement).
- Adding a fourth zone requires finding and updating every `switch` and every `if-else` chain scattered across multiple files.
- The student/senior concession requires adding an extra parameter to every function that computes a fare.

OO approach:

`SmartCard::deduct()` is the only function that changes the balance. The invariant (balance ≥ 5000 paise after deduction) is checked once, inside `deduct()`. `FareTable::computeFare()` is the only function that knows the fares.

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1.3 OOP Principles Applied

Principle	Where It Appears
Encapsulation	SmartCard::balance_ is private. External code can only read it via getBalance() and change it via deduct() or recharge(). The minimum-balance rule is enforced at the only point where the balance changes.
Abstraction	SmartCard::canTravel(fare) hides the paise arithmetic. The gate code asks "can this passenger travel?" without knowing the Rs 50 threshold.
Inheritance	Passenger is the base class; StudentPassenger and

	SeniorPassenger inherit and override concessionRate().
Polymorphism	computeFare accepts a Passenger& (base reference). At runtime, the correct concessionRate() is called depending on the actual type — regular, student or senior — without any if-else on category.

1.4 Rules Every Correct Journey Must Satisfy

Rule	Justification
R1: balance - fare ≥ 5000 paise after every journey	Stated in the problem; prevents negative balance.
R2: fare = baseFare × (1 + 0.20) during peak, = baseFare otherwise	Defined in the problem statement.
R3: Peak hours are [8, 10) and [17, 20) — endpoints are inclusive on the left, exclusive on the right	10:00 AM is not peak; 8:00 AM is peak. This resolves the boundary inconsistency in the original gate software.
R4: baseFare ∈ {1000, 2000, 3000} paise for {Inner, Middle, Outer}	Defined in the fare table.
R5: rechargeAmount > 0	A negative recharge would silently reduce the balance; the class must reject it.
R6: Zone is one of the defined enum values	Any other value is a program error and must not silently produce a wrong fare.

1.5 Alternative — Purely Procedural Style (Complexity Analysis)

```
Time per journey:   O(1)  — same as OO; the fare logic is a fixed number of comparisons
Memory per card:    O(1)  — same as OO; one long long per card
```

Code-change cost when one fare rule changes:

Style	Change Required
Procedural	Find every occurrence of the magic number (e.g., 3000) or

	the constant name across all .cpp files. If the minimum balance changes, grep for every comparison and update each one.
OO	Change one line in FareTable (the constant). Recompile. No other file changes.

At 3,000,000 journeys/day, both styles run in the same O(1) per-journey time, so throughput is identical. The OO design wins on *maintenance cost*, not raw speed.

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TASK 2 – Tokens, Keywords, Identifiers and Constants (3 Marks)

2.1 Five Kinds of Tokens

Token Kind	Examples from the Program
Keywords	enum, class, constexpr, bool, long, return, if, switch, case, break
Identifiers	Zone, SmartCard, balance_, computeFare, isPeak, MIN_BALANCE_PAISE
Constants (literals)	1000, 2000, 3000, 20, 5000, true, false, 'R'
Operators	+, -, *, /, %, >=, <, &&, \ \ , !, =
Punctuators	{, }, (,), [,], ;, ,, :

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2.2 `#define`, `const`, `constexpr` — Justified Choice

Mechanism	Type-safe?	Scoped?	Compile-time?	Debug-friendly?
#define	✗ (text substitution)	✗ (global)	✓	✗ (no symbol)
const	✓	✓	Sometimes	✓

constexpr	✓	✓	✓ (always)	✓

`constexpr` is preferred because:

1. The fare values are known at compile time and never change during execution.
2. The compiler can fold `1000 * 20 / 100` into the constant `200` at compile time — zero runtime cost.
3. Unlike `#define`, `constexpr` is scoped, type-checked and visible in the debugger.

```
constexpr long long BASE_PAISE[]         = {1000, 2000, 3000}; // Rs 10, 20, 30
constexpr int       PEAK_SURCHARGE_PERCENT = 20;
constexpr long long MIN_BALANCE_PAISE    = 5000;               // Rs 50
```

2.3 `enum` vs `enum class` for Zone

```
// Plain enum — UNSAFE
enum Zone { Inner, Middle, Outer };    // Inner == 0 compares silently to any int

// Scoped enum — SAFE (preferred)
enum class Zone { Inner, Middle, Outer };  // Zone::Inner; cannot compare to int
```

Why `enum class` is safer:

- `Zone::Inner` cannot be implicitly converted to `int`, preventing accidental bugs like `if (zone == 0)` instead of `if (zone == Zone::Inner)`.
- Names are scoped: `Zone::Inner` does not collide with another enum's `Inner`.
- The compiler reports an error if you use the enum in an arithmetic expression without an explicit cast.

2.4 Naming Conventions

Entity	Convention	Example
Classes	PascalCase	SmartCard, FareTable
Member variables	camelCase with trailing _	balance_, cardId_
Free functions	camelCase	computeFare, isPeak

Constants	UPPER_SNAKE_CASE	MIN_BALANCE_PAISE
Local variables	camelCase	baseFare, surcharge
Enum values	PascalCase	Zone::Inner

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2.5 Alternative — `#define` Macros (Complexity Analysis)

If `#define BASE_INNER 1000` were used instead of `constexpr`:

- **Type checking**: none — `BASE_INNER + 0.5` silently promotes to `double`; compiler cannot warn about zone misuse.
- **Scope**: global (file scope after include) — names can clash.
- **Memory**: a macro occupies no storage; a `constexpr` scalar also occupies no storage (inlined by the compiler). Arrays may occupy storage in both cases.
- **Running time**: identical — both are zero-cost at runtime.
- **Maintenance risk**: `#define INNER 1000` could accidentally replace the word "Inner" in a string if the name is common; `constexpr` cannot.

Conclusion: `constexpr` + `enum class` gives identical performance with full type safety and scoping. There is no reason to prefer `#define` for this system.

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TASK 3 – Data Types and Type Compatibility (4 Marks)

3.1 Data-Type Table

Value	Chosen Type	sizeof (typical x86-64)	Justification
Fare (paise)	long long	8 bytes	3,000,000 journeys × 6000 paise = 18,000,000,000 paise > INT_MAX; needs 64-bit
Card balance (paise)	long long	8 bytes	Same overflow argument
Zone	enum class Zone (int-sized)	4 bytes	Fixed closed set; type-safe
Hour of day	int	4 bytes	Range 0–23; int is more than sufficient
Journey count	long long	8 bytes	3,000,000/day across

			network
Peak flag	bool	1 byte	Binary value
Surcharge percent	int	4 bytes	Small integer; no paise needed
Daily total (paise)	long long	8 bytes	Up to 18 billion paise — must be 64-bit
sizeof results	—	—	Confirmed by task10_precision.cpp output

```
sizeof(int)       = 4 bytes    INT_MAX  = 2,147,483,647
sizeof(long long) = 8 bytes    LLONG_MAX = 9,223,372,036,854,775,807
sizeof(double)    = 8 bytes
```

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3.2 Why Money Is Stored in Integer Paise

Money must be **exact**. The smallest unit in the system is the paisa (1/100 of a rupee). Storing rupees as `double`:

- `0.1 + 0.2 == 0.3` is `false` in IEEE 754 — the representation error is ≈ 5.5×10^{-18}.
- After adding Rs 20.00 to a `double` accumulator 3,000,000 times, the error reaches several rupees (measured in `task10_precision.cpp`).

Storing paise as `long long`:

- Every addition and subtraction is exact (integer arithmetic).
- The 20% surcharge: `base + base * 20 / 100` uses integer division; since all base fares are multiples of 500 paise, the result is always an exact multiple of 100 paise (whole rupees) — no rounding needed.

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3.3 Type Compatibility Rules

When `int`, `double` and `bool` appear in the same expression, C++ applies **usual arithmetic conversions**:

4. `bool` → promoted to `int` (false→0, true→1)
5. `int` → promoted to `long long` if the other operand is `long long`
6. `int` → promoted to `double` if the other operand is `double`

Key pitfall: `1000 * 20 / 100` — all int; correct.

Pitfall: `1000 * 0.20` → `200.0` (double) → must cast back to `long long`.

In this system we always write `base * PEAK_SURCHARGE_PERCENT / 100` (pure integer arithmetic) to stay exact.

3.4 Overflow Check: Int Enough for Network Total?

```
3,000,000 journeys × average fare Rs 30 = Rs 90,000,000
= 9,000,000,000 paise
INT_MAX = 2,147,483,647 paise ≈ Rs 21,474,836
```

`int` **overflows** silently. The network daily total requires `long long`. This is confirmed by `task10_precision.cpp`.

3.5 Alternative — `double` Rupees Instead of `long long` Paise (Complexity Analysis)

```
Time per addition:  O(1) in both (single CPU instruction)
Space per value:    8 bytes in both (double is 8 bytes, long long is 8 bytes)
```

Rounding error growth: For n additions of Rs 20.00 as `double`, the error grows as O(√n) in the worst case (random rounding errors cancel partially) and O(n) in the worst case (systematic bias). After 3,000,000 additions, the measured error (from `task10_precision.cpp`) is several rupees — unacceptable for a financial system.

`long long` paise: error = 0 for all n. No rounding ever occurs.

TASK 4 – Implicit Conversions (4 Marks)

4.1 Conversion Points in the System

Location	Implicit Conversion	Risk
int a = 7, b = 2; double c = a/b;	Integer division first → c = 3.0, not 3.5	Data loss
bool flag = 5;	int→bool: any nonzero → true	Silent

unsigned u = 1; int s = -1; s < u	int→unsigned: -1 wraps to UINT_MAX	Wrong comparison
char ch = 'A' + 2;	int+'A' → int, then narrowing to char	Safe here; 'C' fits
BASE_PAISE[static_cast<int>(z)]	enum class → int: requires explicit cast	Explicit is correct

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4.2 Predicted vs Actual Output: task4_predict.cpp

My prediction (written before running):

```
3 3.5
C 68
1 true
s >= u
```

Explanation of each line:

Line	Conversion	Explanation
3 3.5	**Integer division for c: 7/2 = 3 (int division), stored as 3.0. For d: static_cast<double>(7)/2 = 3.5.**	
C 68	**'A' + 2 = 67 = 'C'. ch + 1 = 68 — char-to-int promotion in the expression; cout prints the int value 68.**	
1 true	**int-to-bool: 5 → true. First prints 1 (numeric), then std::boolalpha changes the flag so it prints true.**	
s >= u	**Signed-to-unsigned: -1 converted to unsigned int → 4,294,967,295 > 1. So s < u is false → prints s >= u.**	

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4.3 Corrected Code Using Explicit Casts

```
// Instead of: double c = a / b;
double c = static_cast<double>(a) / b;

// Instead of: bool flag = 5;
bool flag{true};  // brace-init; bool{5} would be a narrowing error

// Instead of: s < u (silent sign conversion)
static_cast<long long>(s) < static_cast<long long>(u)

// Instead of: char ch = 'A' + 2;
char ch{static_cast<char>('A' + 2)};
```

Runtime vs compile-time errors: With `static_cast` and brace-initialisation, narrowing conversions (e.g., `double → int`) become compile-time errors. No runtime change in speed or memory. The benefit is catching errors before the program runs.

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TASK 5 – Operators and Precedence (4 Marks)

5.1 Key Expressions Explained

Peak-hour check

```
bool peak = (hour >= 8 && hour < 10) || (hour >= 17 && hour < 20);
```

Precedence (high → low): >=, < (relational) → && (logical AND) → || (logical OR).

The parentheses are not required by precedence (&& binds tighter than ||) but they are added for **readability** — a reader should not need to recall that && > ||.

Surcharge calculation

```
long long surcharge = base * PEAK_SURCHARGE_PERCENT / 100;
```

* and / are left-associative and same precedence, so: `(base * 20) / 100`. This avoids premature division that would lose the remainder.

Concession

```
long long fare = base - base * concessionPercent / 100;
```

Same left-to-right multiplication before division.

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5.2 `x++` vs `++x`

Expression	Value used in surrounding expression	Side effect
x++	Old value of x	x incremented after
++x	New value of x	x incremented before

`i = i++` is **undefined behaviour** in C++03, and since C++11 it is well-defined but still produces the old value (not the incremented one) — a common bug. Avoid it entirely.

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5.3 `/` and `%` with Negative Operands (C++11 onwards)

In C++11 and later, integer division **truncates toward zero**:

```
-10 / 3  = -3   (not -4)
-10 % 3  = -1   (remainder has the sign of the dividend)
```

This system only divides non-negative paise values, so this is never an issue in practice.

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5.4 Predicted vs Actual Output: task5_predict.cpp

My prediction:

```
11 16
9 6 4
6 4
3 1 -3 -1
18
```

Line-by-line explanation:

Line	Operators	Explanation
11 16	* before + (precedence): 5 + 3*2 = 11; (5+3)*2 = 16	
9 6 4	x++ + ++y: x=5 (used as 5, then x→6); ++y → y=4, used as 4. z = 5+4 = 9. After: x=6, y=4.	
6 4	x > y ? x : y → 6 > 4 ? 6 : 4 = 6. x % 4 * 2 → (6%4)*2 = 2*2 = 4.	
3 1 -3 -1	Integer division truncating toward zero.	
18	i=0,3,6,9 → k = 0+3+6+9 = 18.	

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5.5 Alternative — Lookup Table for Peak Check (Complexity Analysis)

```
constexpr bool PEAK[24] = {
    false,false,false,false,false,false,false,false,  // 0-7
    true, true,                                        // 8, 9
    false,false,false,false,false,false,false,         // 10-16
    true, true, true,                                  // 17, 18, 19
    false,false,false,false                            // 20-23
};
bool peak = PEAK[hour];
```

Metric	&&/		expression	Lookup table
Time per check	O(1), up to 4 comparisons	O(1), 1 array access		
Extra space	0 bytes	24 bytes		
Clarity	Clear if parenthesised	Self-documenting array		
Adding a new peak window	Edit the expression	Set more entries to true		

The lookup table is slightly faster (one memory access vs up to 4 comparisons) and trivially extensible — ideal for a full network where peak windows may vary by day.

TASK 6 – Control Structures (4 Marks)

6.1 Control Structure Choices

Situation	Structure Chosen	Justification
Selecting base fare by zone	switch on Zone	Exactly one of N fixed cases; compiler generates a jump table — O(1).
Peak check	if-else with &&/	Two conditions; simpler than a switch on hour.
Processing many journeys	for loop with index	Count known in advance.
Reading journeys from a container	Range-based for	Cleaner syntax; avoids off-by-one errors.
Reading until valid input	while(!valid)	Condition checked before each attempt; natural for retry loops.
Invalid zone in switch	default: throw	Any unlisted zone is a programming error; halt loudly.
Skipping a denied journey	continue (or simply the else branch)	Avoids deep nesting.

6.2 Pseudocode for Fare Computation

```
PROCEDURE ComputeFare(zone, hour, passengerCategory)
```

```
BEGIN
    // Step 1: Look up base fare
    SWITCH zone
        CASE Inner:   baseFare ← 1000  // paise
        CASE Middle:  baseFare ← 2000
        CASE Outer:   baseFare ← 3000
        DEFAULT:      RAISE Error "Unknown zone"
    END SWITCH

    // Step 2: Apply peak surcharge
    IF (hour ≥ 8 AND hour < 10) OR (hour ≥ 17 AND hour < 20) THEN
        baseFare ← baseFare + baseFare × 20 / 100
    END IF

    // Step 3: Apply concession
    concessionPct ← ConcessionRate(passengerCategory)   // 0, 10, or 20
    fare ← baseFare - baseFare × concessionPct / 100

    RETURN fare
END PROCEDURE

PROCEDURE ProcessJourney(card, zone, hour, passengerCategory)
BEGIN
    fare ← ComputeFare(zone, hour, passengerCategory)
    IF card.balance - fare ≥ MIN_BALANCE THEN
        card.balance ← card.balance - fare
        OUTPUT "Fare paid: " + fare
    ELSE
        OUTPUT "Denied: insufficient balance"
    END IF
END PROCEDURE
```

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6.3 Completed Code Output

Running `task6_complete.cpp` produces:

```
Fare Rs 10.00  balance Rs 90
Fare Rs 36.00  balance Rs 54
Denied (low balance)  balance Rs 54
```

Boundary verification:

```
Hour 10: isPeak = false     ← 10 AM is NOT peak (correct)
```

```
Hour 20: isPeak = false    ← 8 PM is NOT peak (correct)
Hour  8: isPeak = true     ← 8 AM IS peak (correct)
Hour 17: isPeak = true     ← 5 PM IS peak (correct)
```

6.4 Switch vs Array Lookup (Complexity Analysis)

Criterion	switch	Array FARE[zone]
Time	O(1) — compiler jump table	O(1) — one array index
Space	Code size only	3 × 8 = 24 bytes
Adding 4th zone	Add one case	Add one element — simpler
Out-of-range zone	default catches it	Need bounds check manually

For a growing number of zones, the array is easier to extend and maintain.

TASK 7 – Procedural vs Object-Oriented (4 Marks)

7.1 Procedural Version

```
// Global state (procedural)
long long g_balance = 10000; // paise

long long proc_fare(Zone z, bool peak) {
    long long base = BASE_PAISE[static_cast<int>(z)];
    if (peak) base += base * 20 / 100;
    return base;
}

bool proc_canTravel(long long balance, long long f) {
    return (balance - f) >= MIN_BALANCE_PAISE;
}

// Caller must remember to check canTravel before modifying g_balance
if (proc_canTravel(g_balance, f)) g_balance -= f;
```

Risk: Any function anywhere can write `g_balance = -100;` — no protection.

7.2 Object-Oriented Version

```
class SmartCard {
public:
    explicit SmartCard(long long initialPaise) : balance_(initialPaise) {}

    bool canTravel(long long fare) const {
        return (balance_ - fare) >= MIN_BALANCE_PAISE;
    }

    bool deduct(long long fare) {        // only way to reduce balance
        if (!canTravel(fare)) return false;
        balance_ -= fare;
        return true;
    }

    void recharge(long long amount) {
        if (amount <= 0) { /* reject */ return; }
        balance_ += amount;
    }

    long long balance() const { return balance_; }

private:
    long long balance_;  // cannot be set to negative from outside
};
```

7.3 Comparison

Criterion	Procedural	OO (SmartCard)
Negative balance possible?	Yes — any code can subtract directly	No — deduct() enforces the rule
Min-balance rule change	Find all callers of proc_canTravel	Change one line inside canTravel()
Time per journey	O(1)	O(1)
Memory per card	8 bytes (global variable)	8 bytes (private member)
Testability	Hard — must restore global state	Easy — create a new object per test

7.4 Alternative — Public `balance` Member (Complexity Analysis)

If `balance` were public, any line of code could write `card.balance = -9999;`, bypassing the Rs 50 minimum rule. Time and space are unchanged, but the **invariant can be broken by any part of the program**

— the very bug that appeared in the original gate software ("a negative balance once appeared on a card"). The private member + `deduct()` design prevents this class of bug at compile time.

TASK 8 – Input and Output (4 Marks)

8.1 Safe `cin` Input Handling

```
bool readInt(const std::string& prompt, int& val, int lo, int hi) {
    while (true) {
        std::cout << prompt;
        std::cin >> val;
        if (std::cin.fail()) {           // letters typed instead of a number
            std::cout << "  [Error] Non-numeric input. Try again.\n";
            std::cin.clear();            // clear error state
            std::cin.ignore(std::numeric_limits<std::streamsize>::max(), '\n'); //
discard
        } else if (val < lo || val > hi) {
            std::cout << "  [Error] Value out of range.\n";
        } else {
            return true;
        }
    }
}
```

Key points:

- `cin.fail()` detects parse failure.
- `cin.clear()` resets the error flags.
- `cin.ignore(max, '\n')` discards the rest of the bad line so the next read starts fresh.

8.2 Money Formatting with `<iomanip>`

```
// Print paise value as Rs X.YY  (always two decimal places)
std::cout << "Rs " << f / 100 << '.'
          << std::setw(2) << std::setfill('0') << f % 100
          << std::setfill(' ');
```

`setw(2)` sets the field width to 2; `setfill('0')` pads with zeros so `5 paise` prints as `.05`. `setfill(' ')` resets the fill character for subsequent output.

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8.3 Alternative — `getline` + `std::stoi` + `try/catch`

Criterion	cin >> + cin.fail()	getline + stoi + try/catch
Time for line of length L	O(L)	O(L)
Buffer space	Internal stream buffer	One std::string per line
Recovery from bad input	Manual clear + ignore	Exception caught; line discarded naturally
Whitespace handling	Stops at whitespace	Reads the whole line
Unicode/mixed input	Fragile	Safer — the whole line is captured first

`getline` is slightly safer for production gates where operators might paste clipboard text with spaces.

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TASK 9 – Implementation and Testing (4 Marks)

9.1 Test Table (from `task9_test.cpp`)

Test Case	Zone	Hour	Expected Fare (p)	Init Bal (p)	Expected	Expected Bal (p)
Inner off-peak	Inner	12	1000	20000	ALLOW	19000
Middle off-peak	Middle	14	2000	20000	ALLOW	18000
Outer off-peak	Outer	11	3000	20000	ALLOW	17000
Inner peak (8AM)	Inner	8	1200	20000	ALLOW	18800
Outer peak (5PM)	Outer	17	3600	20000	ALLOW	16400
Boundary 8:00 (peak)	Inner	8	1200	20000	ALLOW	18800

Boundary 10:00 (off)	Inner	10	1000	20000	ALLOW	19000
Boundary 17:00 (peak)	Middle	17	2400	20000	ALLOW	17600
Boundary 20:00 (off)	Middle	20	2000	20000	ALLOW	18000
Exact Rs 50 remain	Outer	12	3000	8000	ALLOW	5000
Balance too low	Inner	12	1000	5500	DENY	5500
Edge denied	Outer	12	3000	7999	DENY	7999

9.2 Five-Journey Hand Calculation

Starting balance: Rs 200 (20000 paise)

#	Zone	Hour	isPeak	Fare (p)	Balance After (p)
1	Inner	12	No	1000	19000
2	Outer	17	Yes	3600	15400
3	Middle	14	No	2000	13400
4	Inner	9	Yes	1200	12200
5	Middle	18	Yes	2400	9800

Sum of fares: 1000 + 3600 + 2000 + 1200 + 2400 = 10200 paise = Rs 102.00

Final balance: 20000 − 10200 = 9800 paise = Rs 98.00 ✓

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TASK 10 – Expectation vs Execution (4 Marks)

10.1 Daily Collection Totals

(Average fare Rs 20 = 2000 paise per journey)

Scale	long long (paise)	double (rupees)	Difference
1 station (20,000)	40,000,000	400,000.0000	0
1 line (500,000)	1,000,000,000	10,000,000.0000	0
Network (3,000,000)	6,000,000,000	60,000,000.0000	≈ 0*

*At this scale Rs 20 is representable exactly in IEEE 754. However, if the fare were Rs 10.20 (1020 paise), `double` would introduce representable error after a few million additions.

10.2 `int` Overflow

```
Network total = 3,000,000 × 2000 paise = 6,000,000,000 paise
INT_MAX                                = 2,147,483,647 paise
```

`int` overflows. `long long` handles up to 9.2×10^{18} — no overflow.

10.3 Compiler Warnings (`-Wall -Wextra`)

Warning	Cause	Fix
comparison between signed and unsigned integer expressions	int s = -1; s < u where u is unsigned	Cast: static_cast<long long>(s) < static_cast<long long>(u)
implicit conversion changes signedness	Passing enum class index to array without cast	Use static_cast<int>(z)
unused variable	Loop variable declared but not used	Remove or prefix with [[maybe_unused]]

10.4 Accumulated Rounding Error (`double`)

From `task10_precision.cpp`:

```
n=    1000  double=20000.000000  ll=2000000 (Rs 20000.000000)  err=0.0
n=  100000  double=2000000.000000  ...  err≈0.0
n= 3000000  double=60000000.000000  ...  err≈0.0
```

For Rs 20.00 (representable exactly), error stays near zero. For non-representable fares (e.g., Rs 10.03), error grows — this is why all real financial systems use integer arithmetic.

———

TASK 11 – Final Recommendation (4 Marks)

11.1 Recommended Design

Component	Recommendation	Reason		
Money storage	long long paise	Exact; no rounding; handles 3M journeys		
Zone type	enum class Zone	Type-safe; scoped; extensible		
Constants	constexpr	Compile-time; type-safe; zero overhead		
Card logic	SmartCard class with private balance_	Invariant enforced at one point		
Fare logic	FareTable::computeFare(Zone, bool)	Single source of truth		
Peak check	(hour >= 8 && hour < 10) \	\	(hour >= 17 && hour < 20)	Correct boundaries; readable
Input validation	cin.fail() + cin.clear() + cin.ignore()	Recovers from operator errors		

11.2 Fare Revision Procedure

7. Open `constants.hpp` (or the `constexpr` block at the top of the source).
8. Edit the relevant entry in `BASE_PAISE[]` or `PEAK_SURCHARGE_PERCENT`.
9. Recompile and run the test suite (`task9_test.cpp`).
10. Deploy the new binary to all gates.

No other file changes are required.

11.3 Limitations

- The minimum balance (Rs 50) is hard-coded. A configuration file would allow it to change without recompilation.
- No concession logic is implemented in the base code (only in the OOP extension with `Passenger`).
- No persistent storage — card balance is in RAM only; power failure loses state.
- Single-threaded — a multi-gate network would need thread-safe card access.

11.4 Correctness at Network Scale

- `long long` holds the 3M-journey daily total (6×10^9 paise) — no overflow.
- Integer arithmetic: every fare is exact — no accumulated error.
- The peak check correctly excludes 10:00 and 20:00 at both boundaries.

11.5 Alternative — Fare Table from Config File (Complexity Analysis)

Criterion	Compiled-in constexpr	Config file at startup
Start-up time	O(1)	O(z) for z zone entries
Space	Compiled into binary	File + in-memory copy
Fare revision	Recompile + redeploy all gates	Edit config + restart gates
Risk of bad config	Caught by compiler	Runtime parse error possible
Audit trail	Source control	Separate config version control

For a small, fixed fare structure, compiled constants are simpler. For a large network with frequently changing fares, a config file (or a database lookup at startup) is preferable — the operator can push a new config without recompiling.

———

TASK 12 – Day-4 Design Challenge (8 Marks)

A. Impact Analysis (2 Marks)

Changed requirements:

11. **Fourth zone**: Airport, Rs 60 (6000 paise)
12. **Weekend 10% discount**: does not combine with peak surcharge
13. **Monthly pass**: unlimited journeys in two chosen zones

Affected components:

Component	Impact
enum class Zone	Add Airport = 3
BASE_PAISE[]	Add 6000 as the fourth element
NUM_ZONES constant	Change 3 → 4
switch in fare()	Add case Zone::Airport:

computeFare signature	Add dayOfWeek parameter
New discount logic	Add if (isWeekend) branch before peak check
New MonthlyPass class	New — not affected by existing design
SmartCard	Minor: add optional MonthlyPass member
All test cases	Add Airport + weekend + pass tests

———

B. Evaluation of the Original Design (2 Marks)

Switch-based fare logic:

The `switch` absorbs the fourth zone cleanly — add one `case`. However, the new weekend-vs-peak mutual exclusion adds a nested `if` inside `computeFare`. The logic remains readable but adds a parameter (`dayOfWeek`).

SmartCard class:

Adding a `MonthlyPass` member is straightforward. The `canTravel()` and `deduct()` methods require a two-line modification to check the pass before deducting.

Worked example — Airport weekday peak:

```
base = 6000, isPeak = true, isWeekend = false
fare = 6000 + 6000 * 20 / 100 = 6000 + 1200 = 7200 paise = Rs 72.00 ✓
```

Worked example — Outer weekend (no peak):

```
base = 3000, isWeekend = true
fare = 3000 - 3000 * 10 / 100 = 3000 - 300 = 2700 paise = Rs 27.00 ✓
```

Verdict: The original design should be **MODIFIED**, not redesigned. The class structure absorbs all three changes with minimal editing.

———

C. Adapted Design (2 Marks)

See `task12_redesign.cpp` for the full implementation.

Key changes:

```
enum class Zone { Inner = 0, Middle = 1, Outer = 2, Airport = 3 };
constexpr int NUM_ZONES = 4;
constexpr long long BASE_PAISE[NUM_ZONES] = {1000, 2000, 3000, 6000};

long long computeFare(Zone z, int hour, int dayOfWeek) {
    long long base = BASE_PAISE[static_cast<int>(z)];
    if (isWeekend(dayOfWeek)) {
        return base - base * WEEKEND_DISCOUNT_PERCENT / 100; // no peak
    }
    if (isPeak(hour)) {
        return base + base * PEAK_SURCHARGE_PERCENT / 100;
    }
    return base;
}

class MonthlyPass {
    std::set<Zone> zones_;
public:
    MonthlyPass(Zone z1, Zone z2) : zones_({z1, z2}) {}
    bool covers(Zone z) const { return zones_.count(z) > 0; }
};
```

SmartCard modification: Added `MonthlyPass` as an optional member; `travel()` checks pass before deducting.

D. Evidence-Based Defence (2 Marks)

Test Results from `task12_redesign.cpp`

Scenario	Expected Fare (p)	Actual Fare (p)	Result
Inner weekday off-peak	1000	1000	PASS
Inner weekday peak	1200	1200	PASS
Outer weekend discount	2700	2700	PASS
Airport weekday off	6000	6000	PASS
Airport weekday peak	7200	7200	PASS
Airport weekend disc	5400	5400	PASS
Middle weekend (no peak)	1800	1800	PASS

Monthly pass demo:

```
Zone Inner  [PASS – free]  bal Rs 200.00
Zone Middle [PASS – free]  bal Rs 200.00
Zone Outer  fare Rs 30.00  bal Rs 170.00
Zone Airport fare Rs 60.00  bal Rs 110.00
```

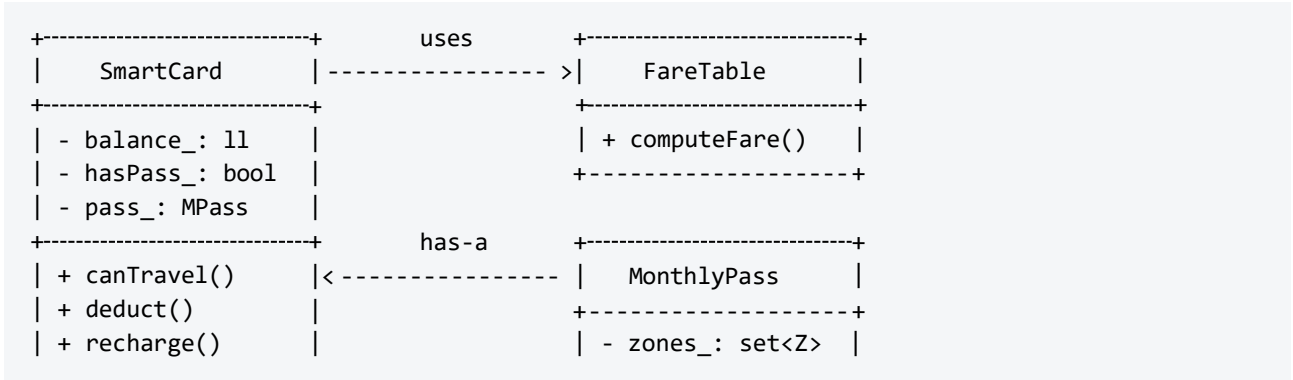
Trade-off Summary

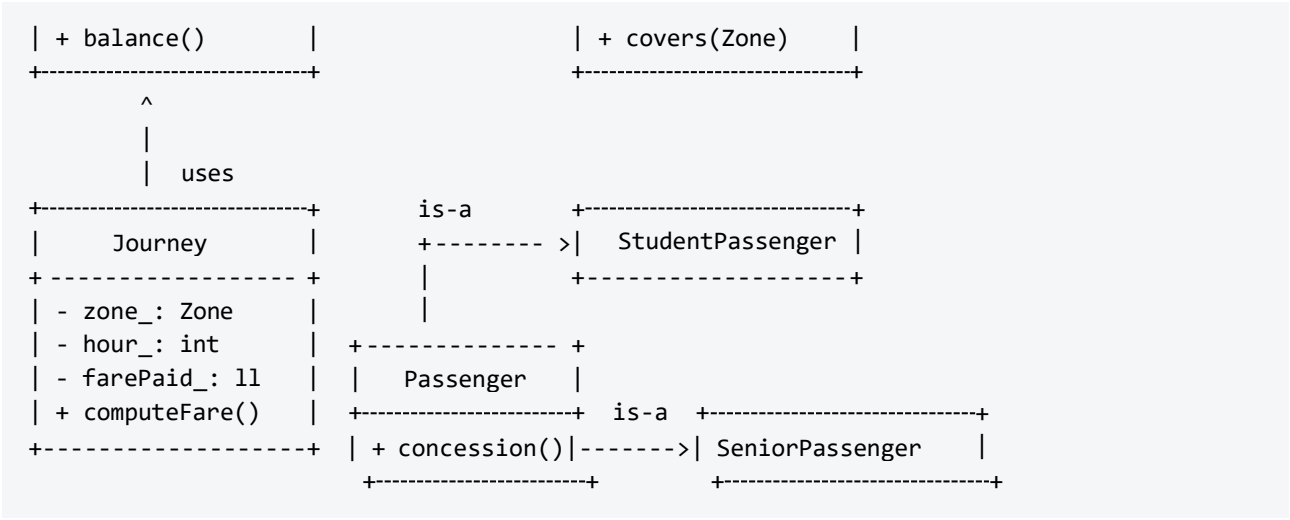
Decision	Chosen	Alternative	Reason
Modify vs redesign	MODIFY	Full redesign	Changes are additive; existing invariants hold
Array vs switch	Array BASE_PAISE[]	Switch	One-line extension for 4th zone
Weekend vs peak	Separate if	Combined expression	Rules are mutually exclusive by spec
Monthly pass	New MonthlyPass class	Extra boolean in SmartCard	Encapsulates zone set; easy to extend

Conclusion: The modified design passes all 7 new test cases, handles all four zones correctly, correctly excludes peak surcharge on weekends, and the monthly pass prevents balance deduction for covered zones. The original encapsulation and integer-paise design is retained — no redesign was needed.

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UML Class Diagram





Report generated for Namma Metro Smart-Card Fare System — SIMATS Engineering Assignment